Design and Analysis of Micro-strip Patch Antenna for Wearable Devices Using Different Substrates

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***Abstract*—Wearable antennas play an important role in health monitoring systems by enabling global communication and tracking. To make wearable antennas effective, factors as flexibility, safety, human body compatibilities and data transmission rate must be considered. This study focuses on designing and simulating a ground-slotted rectangular microstrip patch antenna using CST Studio Suite, analyzing its performance with various flexible substrates. Materials such as Polyethylene, RT Duroid, Teflon, and polydimethylsiloxane (PDMS) were used as substrates to maintain an operating frequency between 3.5 and 3.8 GHz. Key parameters like reflection coefficient, radiation pattern and gain were evaluated while keeping all substrates at the same height.**

I. Introduction

Wearable microstrip patch antennas have emerged as a crucial technology in the rapidly evolving field of wireless body area networks (WBANs) and wearable electronics. These antennas are designed to be flexible, lightweight, and compact, making them suitable for integration into clothing or direct application on the human body. [1] [2].

A recent study showcased the design of a stretchable microstrip antenna operating at 3.5 GHz, utilizing a polydimethylsiloxane (PDMS) composite substrate to ensure the flexibility required for wearable medical devices. [3]. The 3.5 GHz band enables higher data transmission speeds and lower latency compared to current 4G networks, allowing for real-time monitoring of vital signs and improved patient care [4]. Additionally, the compact size achievable at this frequency is crucial for integration into wearable healthcare devices, ensuring patient comfort while maintaining reliable wireless connectivity for continuous health monitoring [5]. The use of flexible substrates like PDMS composites also offers potential for on-body wearable electronics that can easily bend with human body movements [3], making these antennas particularly suitable for long-term health monitoring applications.

A study presented a dual-band antenna designed for 5G and Wi-Fi applications. When mounted near a human arm model, the antenna exhibited SAR values of 0.370 W/kg at 3.5 GHz and 0.870 W/kg at 5.8 GHz. These values are below the FCC threshold of 1.6 W/kg, indicating that the antenna design meets safety standards for human exposure [14]. 1.6 W/kg averaged over 1 gram of tissue for public exposure from cellular telephones is the safe (FCC) Specific Absorbtion rate (SAR).

A paper discussed a low-profile wearable microstrip patch antenna operating at 2.45 GHz for constant monitoring of vital signs, using polyester textile fabric as a flexible substrate [1]. The antenna achieved a return loss of -10.52 dB and a gain of 7.81 dB. Another study investigated bandwidth enhancement techniques for a wearable microstrip patch antenna operating at 2.3 GHz, utilizing multiple slots on the patch and a complementary rhombus resonator (CRR) on the ground plane [6]. This approach increased bandwidth by 78% compared to antennas with only slots on the patch. The purpose of this study is to evaluate the antenna parameters using different substrates and to discuss which substrate is the most user-friendly, sufficiently flexible, and offers better gain.

II. Antenna Design

Microstrip patch antenna finds applications across both medical and non-medical fields. While it has some drawbacks, such as limited bandwidth and higher copper loss at elevated frequencies, it also offers significant advantages, including being lightweight, easy to print on circuit boards, having a low-profile planar design, and being cost-effective to fabricate.

Initially, the standard equations for designing a microstrip patch antenna were used, resulting in a larger antenna. However, since the system required a compact antenna, the design was revisited and modified to make it suitable for on-body devices. A compact size is essential for the development of wearable devices, making the micro-strip patch antenna a suitable choice. The antenna design consists of three layers: the ground plane, the substrate, and the antenna itself. Radiation occurs primarily from two sides of the antenna rather than from all four sides. When the antenna is placed in close proximity to the human body, its resonant frequency shifts compared to its behavior in free space. This results in significant de-tuning towards lower frequencies, caused by the high dielectric constant of human body tissue [7]. Additionally, the lossy nature of the human body causes it to absorb some of the radiated power, leading to a reduction in the antenna's gain [8]. To determine the patch's width and length, the following parameters are considered.

$$\epsilon_{reff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2}[1 + \frac{12h}{w}]^{\frac{-1}{2}} \quad (1)$$

Since the fringing field around the radiating patch extends beyond the substrate, the dielectric constant is higher than the effective dielectric constant. The key factor determining the operationg frequency is the patch length. The effective length and the width of the antenna is [8]:

$$l_{reff} = \frac{c}{2f_0\sqrt{\epsilon\epsilon_{reff}}} - 2dl \quad (2)$$

Here, dl represents the extended length, which results from the fringing field, and is expressed as

$$(dl) = 0.412h\frac{(\epsilon_{reff} + 0.3)(\frac{w}{h} + 0.264)}{(\epsilon_{reff} - 0.258)(\frac{w}{h} + 0.8)} \quad (3)$$

$$W = \frac{C}{2f_0\sqrt{\frac{(\epsilon_r+1)}{2}}} \quad (4)$$

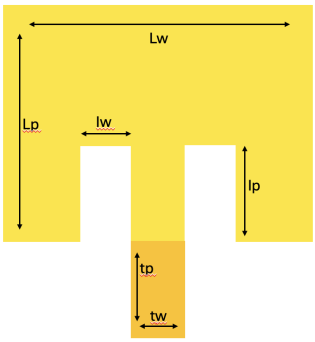


Fig. 1. Fundamental geometric framework of the antenna using PDMS as Substrate.

TABLE I
ANTENNA'S DIMENSION (MM)

Lw	Lp	Iw	Ip	tw	tp
30	26.8	5.1	8	5	10

TABLE II
GROUND (ANTENNA) DIMENSION (MM)

L	W	s	n	t
40	40	18	16	12

Figures 1 and 2 illustrate the dimensions of the antenna and the ground plane. The ground plane is slotted. A study in 2014 showed that multiple slots in the ground plane could enhance the bandwidth of a rectangular patch antenna [10]. Another study demonstrated bandwidth enhancement from 47 MHz to 540 MHz (approximately 493 MHz increase) by adding vertical slots in the ground plane, while maintaining similar

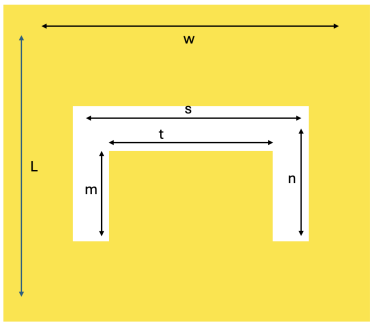


Fig. 2. Copper ground with a U-shaped slot.

gain and directivity to the conventional antenna [11]. Tables 1 and 2 provide the approximate numerical values for the dimensions of the antenna.

Microstrip line feed technique is used for impedance matching at 50 Ω.

TABLE III
DIELECTRIC PROPERTIES OF FLEXIBLE SUBSTRATES

List of Substrates	Properties	
	Dielectric constant er	*Loss tangent*
1. PDMS	2.8	0.02
2. Polyethylene	2.25	0.001
3. RT Duroid	2.2	0.0009
4. Teflon	2.1	0.001

III. SIMULATION RESULT

In this study, rectangular microstrip patch antennas with a resonant frequency of 3.5 GHz were designed on a flexible substrate using the CST Studio Suite Learning Edition. The gain, radiation pattern for phi = 0 and phi = 90 is analyzed in conjunction with reflection coefficient ($\leq$ 10).

Return loss represents the power lost in the load and is used to characterize the input-output behavior of a signal source [12]. The reflection coefficient graph is shown in Figure 3, where PDMS demonstrates the best return loss of -21 dB compared to Polyethylene, RT Duroid, and Teflon. A higher return loss value indicates a better impedance match, bringing it closer to the desired impedance. TABLE III demonstrates PDMS substrate has the highest loss tangent 0.02. To address this, low-profile (smooth) copper is required to ensure a strong bond between the copper patch and the substrate. This configuration results in maximum return loss due to the skin effect in conductors.

An antenna with a low dielectric permittivity substrate has lower return loss and higher bandwidth because the material works as a good insulator and prevents electrons from moving into it [15], can be observed from Fig 3. The 3D polar plot shown in Figure 4,5,6,7 illustrate the gain of the radiated signal and the intensity in different directions. PDMS achieves the

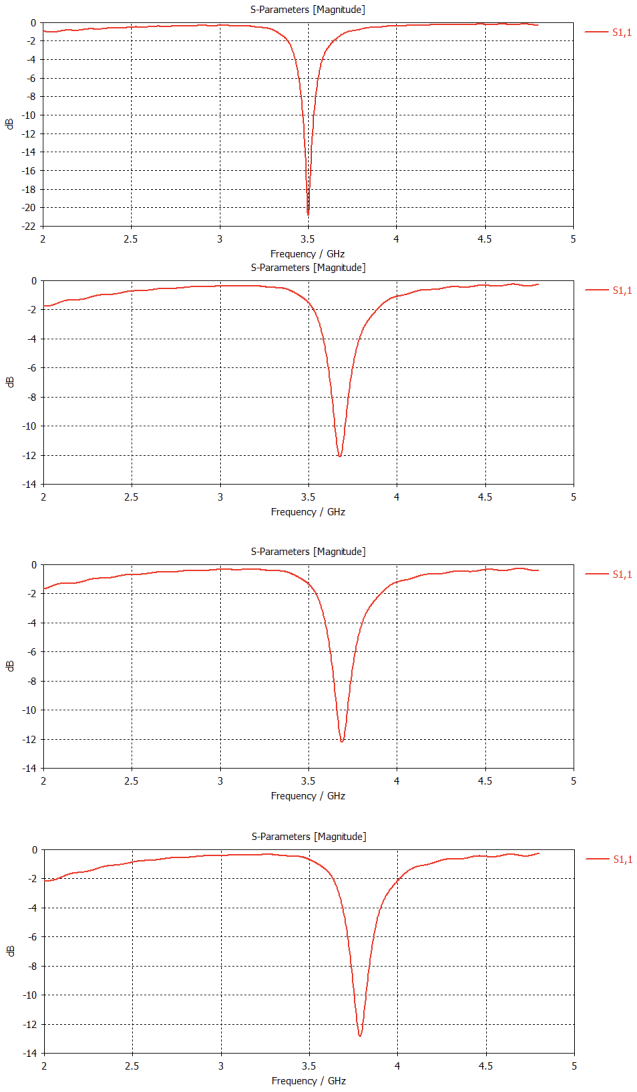


Fig. 3. Reflection coefficient versus frequency graph (a) PDMS (b) Polyethylene (c) RT Duroid (d) Teflon.

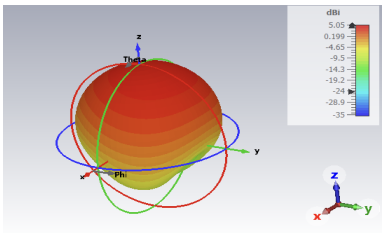


Fig. 4. 2D polar plot for PDMS antenna

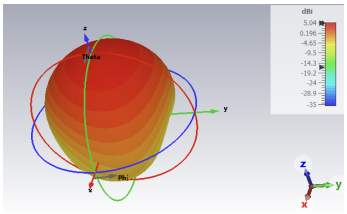


Fig. 5. 2D polar plot for Polyethylene antenna

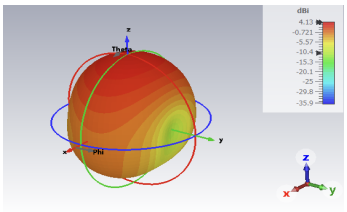


Fig. 6. 2D polar plot for RT duroid antenna

highest gain, measured at 5.05 dBi. the gain can be improved by using different slot designs and adjusting the substrate height in the antenna. Increasing the height (thickness) of the substrate can also lead to improved gain. This is because a thicker substrate generally allows for better radiation efficiency.

VSWR value below 2 is considered acceptable for most antenna applications, including wearable antennas. For sensitive applications, it is expected that the VSWR value remains within 1.5. The simulation result for this antennas are within the desired range.

The human body can influence the radiation patterns [16]. In this work, radiation pattern demonstrates omnidirectional characteristics for this flexible substrate. PDMS, Polyethylene, and RT Duroid exhibit a more omnidirectional pattern, which is ideal for wearable devices as it enhances on-body communication. C0-polarization and cross-polarization are defined for the radiating E-plane and H-plane at fig. 8,9,10,11. The performance of all antennas with various flexible substrates is summarized in Table IV. A comparison of various materials was conducted in this work. The objective of the design is to identify the most suitable substrate for a wearable antenna.

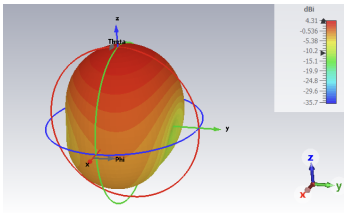


Fig. 7. 2D polar plot for Teflon antenna

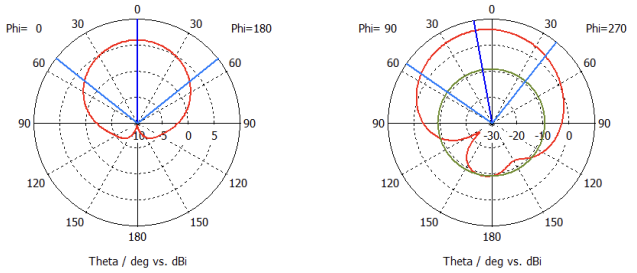


Fig. 8. Radiation Pattern of PDMS at phi=0 and phi=90.

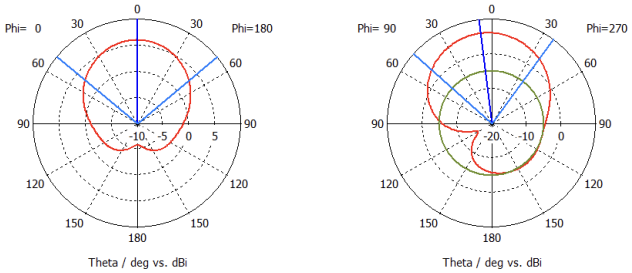


Fig. 9. Radiation Pattern of Polyethylene at phi=0 and phi=90.

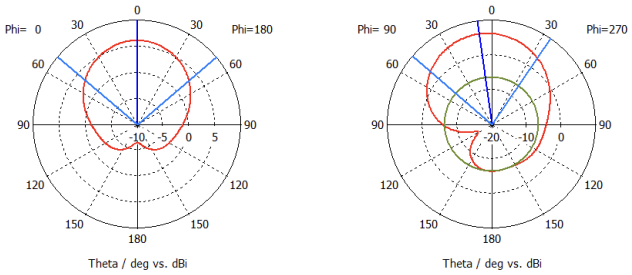


Fig. 10. Radiation Pattern of RT duroid at phi=0 and phi=90.

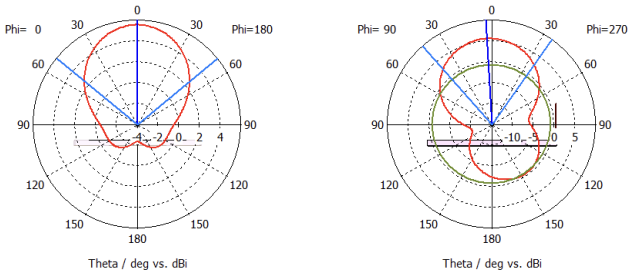


Fig. 11. Radiation Pattern of Teflon at phi=0 and phi=90.

IV. Conclusions

For wearable devices, flexibility is prioritized for comfortable use. PDMS provides the required flexibility while offering excellent return loss, easy impedance matching, and improved gain in comparable to Polyethylene, RT duroid, Teflon. The simulation results indicate that PDMS is the most appropriate substrate due to its well-balanced dielectric constant and loss tangent. However, further research is needed to assess the antenna's compatibility with the human body and to evaluate its Specific Absorption Rate (SAR) levels.

TABLE IV
Performance Assessment of Diverse Flexible Substrates

List of Substrate	Operating Frequency (GHz)	Return Loss (dB)	Gain (dBi)
PDMS	3.5	-21	5.05
Polyethylene	3.62	-12.3	5.04
RT Duroid	3.67	-12.2	4.13
Teflon	3.74	-12.8	4.31

References

[1] Hussain, S., Hafeez, S., Memon, S.A. and Pirzada, N., 2018. Design of wearable patch antenna for wireless body area networks. International Journal of Advanced Computer Science and Applications, 9(9).

[2] Dhanabalan, S.S., Sitharthan, R., Madurakavi, K., Thirumurugan, A., Rajesh, M., Avaninathan, S.R. and Carrasco, M.F., 2022. Flexible compact system for wearable health monitoring applications. Computers and Electrical Engineering, 102, p.108130.

[3] Rana, M.S., Chakrobortty, P., Pal, S., Rahman, M.M., Sarker, A., Saha, P. and Rahman, S., 2024. A review on microstrip patch antenna for wireless communication systems at 3.5 GHz. Bulletin of Electrical Engineering and Informatics, 13(4), pp.2397-2407.

[4] Sriramulu, D., Saini, G. and Dhaliwal, B.S., 2020. Design of wearable microstrip patch antenna for WBAN applications to enhance bandwidth using multiple slots and CRR. International Journal of Advanced Research in Engineering and Technology (IJARET), 11(8), pp.319-327.

[5] Shakhirul, M.S., Ain, M.F., Ahmad, Z., Abidin, I.S.Z. and Ali, M.Z., 2021, May. Stretch analysis of polydimethylsiloxane composite microstrip patch antenna for 5G application. In AIP Conference Proceedings (Vol. 2339, No. 1). AIP Publishing.

[6] Sriramulu, D., Saini, G. and Dhaliwal, B.S., 2020. Design of wearable microstrip patch antenna for WBAN applications to enhance bandwidth using multiple slots and CRR. International Journal of Advanced Research in Engineering and Technology (IJARET), 11(8), pp.319-327.

[7] Abd Rahman, N.H., Yamada, Y. and Amin Nordin, M.S., 2019. Analysis on the effects of the human body on the performance of electro-textile antennas for wearable monitoring and tracking application. Materials, 12(10), p.1636.

[8] Bernard, V. and Iloh, J.P.I., 2013. Microstrip antenna design using transmission line model. International Journal of Emerging Technology and Advanced Engineering, 3(11), pp.410-415.

[9] Balanis, C.A., 2015. Antenna theory: analysis and design. John wiley & sons.

[10] Chauhan, S. and Singhal, P.K., 2014. Enhancement of bandwidth of rectangular patch antenna using multiple slots in the ground plane. International Journal of Research in Electron. And Comm. Technoly, 1, pp.30-33.

[11] Khraisat, Y.S., 2018. Increasing microstrip patch antenna bandwidth by inserting ground slots. Journal of Electromagnetic Analysis and Applications, 10(1), pp.1-11.

[12] Paul, L.C., Hosain, M.S., Sarker, S., Prio, M.H., Morshed, M. and Sarkar, A.K., 2015. The effect of changing substrate material and thickness on the performance of inset feed microstrip patch antenna. American Journal of Networks and Communications, 4(3), pp.54-58.

[13] Victor, A., Ribeiro, J.E. and Araújo, F.F., 2019. Study of PDMS characterization and its applications in biomedicine: A review. Journal of Mechanical Engineering and Biomechanics, 4(1), pp.1-9.

[14] Renit, C. and Raj, T.A.B., 2024. Wearable frequency selective surface-based compact dual-band antenna for 5G and Wi-Fi applications. Automatika, 65(2), pp.454-462.

[15] Tiercelin, N., Coquet, P., Sauleau, R., Senez, V. and Fujita, H., 2006. Polydimethylsiloxane membranes for millimeter-wave planar ultra flexible antennas. Journal of Micromechanics and Microengineering, 16(11), p.2389.

[16] Zhang, K., Vandenbosch, G.A. and Yan, S., 2020. A novel design approach for compact wearable antennas based on metasurfaces. IEEE Transactions on biomedical circuits and systems, 14(4), pp.918-927.